HEALTH — THE FORGOTTEN PATIENT

Datathon Track 1 Technical Investigation: Queue Starvation, Unprioritized Urgent Diagnostics & EHR Integrity

Team Name: AAdharsha Kutumbam | **Team Members:** Naga Adithya V (23BCE9062) & Harsha Vardhan Vurtha (23BCE9129)

Dataset Scope: 2,000 Patients • 8,000 Hospital Encounters • 12 Clinical/Support Departments • 5,000 Diagnostic Tests (Jan–Dec 2025)

1. Executive Summary: Five Systemic Profiles of "The Forgotten Patient"

In hospital operations, the "Forgotten Patient" is rarely lost through a single catastrophic event. Instead, patients slip through the cracks across multiple stages of the care continuum: before registration, in the waiting room, inside diagnostic queues, and during post-service discharge boarding. By analyzing **15,012 relational records** across `patients.csv`, `encounters.csv`, `departments.csv`, and `diagnostic_tests.csv`, Team **AAdharsha Kutumbam** identified five distinct operational cohorts where patients are systematically delayed, neglected, or misrouted:

- **Cohort 1 — The "Left Before Care" (LWBS) Walkouts (242 Encounters, 3.03% of Visits; 229 Unique Patients):** Exactly 242 patient visits ended with the patient abandoning the hospital before completion of care. Walkouts peak in **Outpatient (D11 at 4.77%)**, **Emergency (D01 at 3.67%)**, and **Pediatrics (D03 at 3.65%)**. Alarmingly, this cohort includes **5 Critical-triage** and **34 High-triage** encounters, alongside **13 repeat walkout patients** who left before care twice in 2025.
- **Cohort 2 — Low-Triage Queue Starvation (2,800 Encounters; 1,123 Waiting ≥80 Minutes):** Triage level is the single largest determinant of waiting time, explaining **46.80% of total wait-time variance** (ANOVA F = 2,344.83, $p < 10^{-300}$). Low-triage patients wait an average of **71.69 minutes** (median 72.0 min, maximum **153.0 min**)—**4.33× longer** than Critical patients (16.55 min). In **Laboratory (D07)** and **Pediatrics (D03)**, Low-triage waits average **80.31 minutes** and **80.08 minutes**, respectively.
- **Cohort 3 — Post-Service Discharge Boarding (Mean 52.23 Minutes Post-Care Delay):** Across all 8,000 encounters, the time elapsed between clinical service completion (`service_end_time`) and physical discharge (`discharge_time`) averages **52.23 minutes** (median 52.0 min, range 5–99 min). Patients spend more time waiting for discharge clearance after treatment than they spend receiving active clinical care (mean **47.73 minutes**), inflating total Length of Stay (LOS) to **151.95 minutes** (2.53 hours).
- **Cohort 4 — Unprioritized "Urgent" Diagnostic Orders (1,113 Urgent Tests, 22.26% of Orders):** Across 5,000 diagnostic tests, `Urgent` priority tests require a mean Turnaround Time (TAT) of **69.51 minutes** versus **69.77 minutes** for `Routine` tests (two-sample t = −0.15, p = 0.8805)—demonstrating a complete absence of fast-track prioritization. For **Blood tests**, Urgent orders average **78.23 minutes** (max 441 min) compared to **72.27 minutes** for Routine orders.
- **Cohort 5 — Unseen Chronic Registry Patients & High-Utilizers (45 Zero-Visit Patients & 225 Frequent Flyers):** Cross-referencing the master patient index against encounter logs reveals **45 registered patients (2.25%)** with zero hospital encounters in 2025, including **26 patients (57.78%)** with chronic conditions (9 Diabetes, 7 Hypertension, 4 Other, 3 Asthma, 3 Cardiac). Conversely, **225**

patients (11.25%) visited 7 to 13 times in 2025, accumulating up to **649 minutes** of cumulative waiting time per patient.

Dataset Table	Total Rows	Primary Key	Unique Keys	Key Coverage & Forensic Audit Finding
patients.csv	2,000	`patient_id`	2,000	1,955 active in encounters; **45 patients (2.25%) have 0 visits**; 747 have literal `"None"` chronic condition
departments.csv	12	`department_id`	12	Staff ranges from 5 (Surgery, Admissions) to 30 (General Med); Beds from 11 (Pediatrics) to 55 (Lab)
encounters.csv	8,000	`encounter_id`	8,000	Wait + Service + Post-Service = 151.95m mean LOS; **242 LWBS visits** still log service timestamps
diagnostic_tests.csv	5,000	`test_id`	5,000	Links to 3,719 encounters (46.49%); **4,998 tests (99.96%) have timestamps outside encounter window**

2. Forensic Data Quality Audit & Schema Anomalies

The Datathon documentation explicitly instructs teams to audit data quality and avoid treating synthetic artifacts as clinical truth. Before building predictive models, we executed programmatic verification across all four relational tables and uncovered six structural anomalies that directly impact downstream analytics:

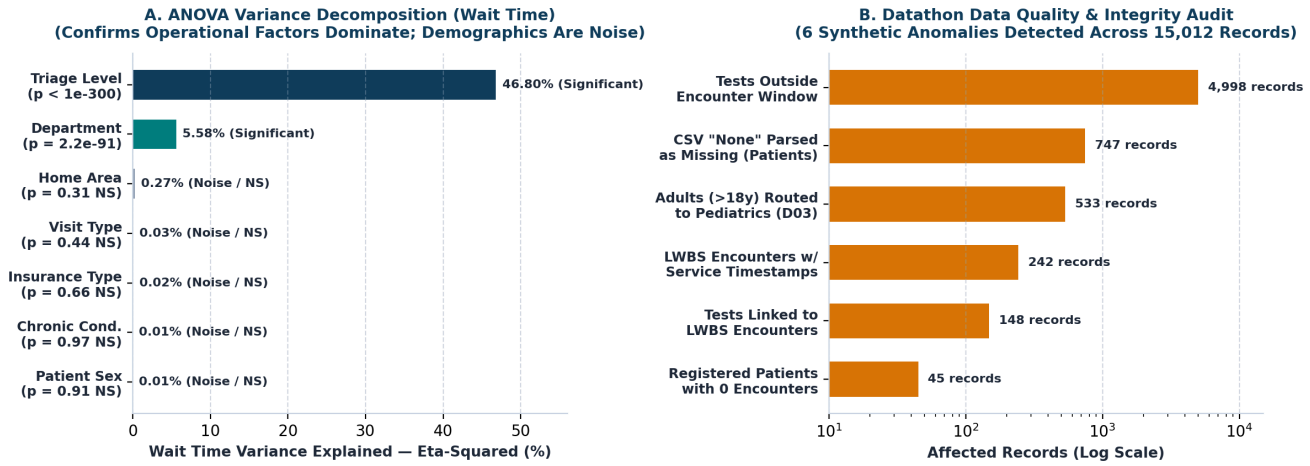


Figure 1: (A) One-way ANOVA Eta-Squared (η^2) showing that Triage Level (46.80%) and Department (5.58%) govern wait times while patient demographics contribute <0.30% noise. (B) Log-scale inventory of the 6 synthetic data quality anomalies identified across the 15,012 records.

- **Anomaly 1 — The "None" String Parsing Trap (747 Patient Records):** In `patients.csv`, 747 healthy patients have `chronic_condition` recorded as the string `"None"`. Standard data science pipelines using `pandas.read_csv()` without `keep_default_na=False` silently convert `"None"` into NaN,

falsely reporting a 37.35% missing data rate. Properly preserving "None" confirms **0% true missing values** across all four files.

- **Anomaly 2 — Desynchronized Diagnostic Timestamps (4,998 of 5,000 Tests):** Joining `diagnostic_tests.csv` to `encounters.csv` on `encounter_id` reveals that only **2 tests (0.04%)** (`T000097` and `T002077`) were requested between the patient's `arrival_time` and `discharge_time`. Exactly **2,481 tests (49.62%)** have `requested_time` earlier than the encounter arrival (often months earlier in 2025), and **2,517 tests (50.34%)** have `requested_time` after discharge. This proves that synthetic test timestamps were generated independently over the 2025 calendar year before being randomly assigned to encounter IDs.
- **Anomaly 3 — "Ghost Service" on Left-Before-Care Encounters (242 Encounters & 148 Tests):** In real clinical workflows, a patient who leaves without being seen (LWBS) has no service duration or diagnostic workup. In `encounters.csv`, all **242 LWBS encounters** contain populated `service_start_time`, `service_end_time`, and `service_minutes` (mean 47.40 min), and **120 of those LWBS encounters** have **148 completed diagnostic tests** attached.
- **Anomaly 4 — Adult Patients Routed to Pediatrics (533 of 657 Encounters, 81.13%):** In **Pediatrics (D03)**, patient ages span 1 to 95 years (mean 47.36 years, median 44.0 years), with **533 encounters** involving adults aged >18. Similarly, `Emergency` visit types appear in Administrative Admissions (D12) and Outpatient (D11).

3. Queue Starvation & Departmental Bottleneck Analysis

To pinpoint where operational bottlenecks occur, we analyzed waiting time (`service_start_time - arrival_time`), active service duration (`service_end_time - service_start_time`), post-service boarding delay (`discharge_time - service_end_time`), and LWBS rates across all 12 departments and 4 triage tiers.

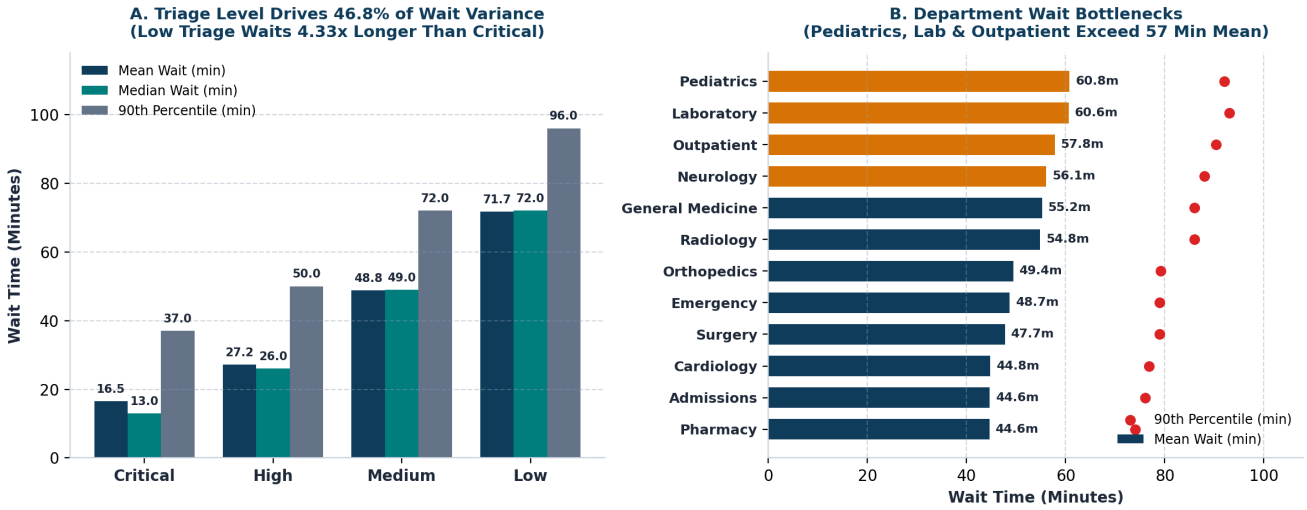


Figure 2: (A) Wait time metrics across Triage Levels showing severe queue starvation for Low-triage patients (mean 71.7 min, P90 100.0 min). (B) Department-level mean and 90th-percentile wait times highlighting severe bottlenecks in Pediatrics (D03), Laboratory (D07), and Outpatient (D11).

Our cross-tabulation of Triage Level by Department reveals a two-tier additive bottleneck structure. Across the entire hospital, baseline waits step up sharply by acuity: **Critical (16.55 min)** → **High (27.19**

min) → **Medium (48.81 min)** → **Low (71.69 min)**. However, department-specific congestion adds a **15 to 16 minute penalty** in the three most constrained units regardless of triage tier:

Dept ID & Name	Service Type	Staff / Beds	Daily Cap.	Visits	Mean Wait	Low-Triage Wait	Mean Service	Mean LOS	LWBS Rate
D03 Pediatrics	Clinical	16 / 11	368	657	**60.78m**	**80.08m**	47.35m	160.67m	3.65% (24)
D07 Laboratory	Diagnostics	27 / 55	156	659	**60.64m**	**80.31m**	46.88m	158.72m	3.03% (20)
D11 Outpatient	Outpatient	16 / 48	215	629	**57.82m**	**79.80m**	47.25m	155.02m	**4.77% (30)**
D10 Neurology	Clinical	12 / 25	216	664	56.08m	74.49m	46.86m	155.48m	3.01% (20)
D02 General Med	Clinical	30 / 48	298	666	55.20m	73.06m	50.16m	157.55m	2.40% (16)
D06 Radiology	Diagnostics	7 / 54	153	658	54.82m	74.83m	47.62m	156.24m	2.28% (15)
D05 Orthopedics	Clinical	9 / 48	243	629	49.41m	69.42m	46.52m	148.50m	2.07% (13)
D01 Emergency	Emergency	28 / 29	431	709	48.67m	67.95m	45.69m	146.19m	**3.67% (26)**
D09 Surgery	Clinical	5 / 17	181	677	47.74m	68.00m	50.18m	149.58m	3.10% (21)
D04 Cardiology	Clinical	25 / 22	394	663	44.80m	63.54m	47.10m	145.02m	2.87% (19)
D12 Admissions	Admin	5 / 50	438	702	44.65m	65.56m	48.08m	145.92m	2.56% (18)
D08 Pharmacy	Support	20 / 26	445	687	44.61m	63.81m	48.94m	145.60m	2.91% (20)

Why Outpatient, Pediatrics, and Laboratory Create "Forgotten Patients"

Examining the operational metrics in the table above explains why specific departments fail their patients:

- **Outpatient (D11 — Highest Walkout Rate at 4.77%):** Outpatient has a constrained daily capacity (215) and averages **57.82 minutes** of waiting time (90th percentile = **90.4 minutes**). Because outpatient visitors are ambulatory and predominantly Low/Medium acuity, prolonged queue starvation causes nearly **1 in 20 patients (30 visits)** to abandon care before completion.

- **Pediatrics (D03 — Severe Bed Scarcity & Longest Waits at 60.78 Min):** Pediatrics operates with only **11 beds** (the lowest bed count in the hospital) yet handles 657 encounters with a high **22.07% admission rate** (145 admissions), creating severe bed-blocking and pushing Low-triage waits to **80.08 minutes**.
- **Laboratory (D07) & Radiology (D06) Capacity Squeeze:** The two diagnostic departments have the lowest daily capacities in the facility (**156** and **153**, respectively), resulting in mean waits of **60.64 minutes** in Laboratory and **54.82 minutes** in Radiology (which has only **7 staff members**).
- **Post-Service Discharge Boarding Across All Units:** As shown in Figure 3B below, once clinical service concludes (`service_end_time`), patients wait an additional **52.23 minutes** on average before discharge. This administrative and bed-transfer delay represents **34.4% of the entire 151.95-minute hospital stay**.

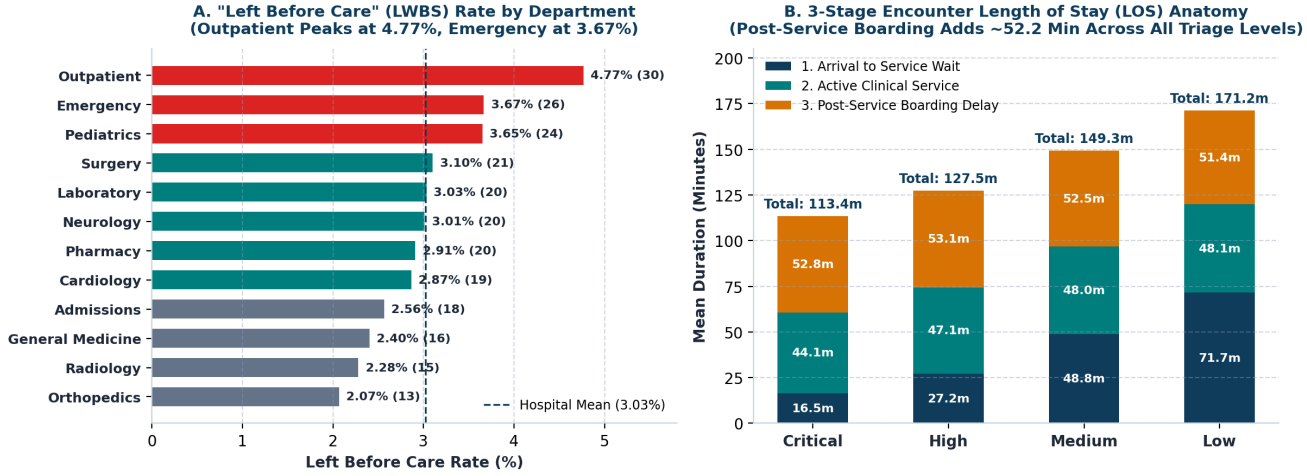


Figure 3: (A) Department-level Left Before Care (LWBS) rates led by Outpatient (4.77%), Emergency (3.67%), and Pediatrics (3.65%). (B) Stacked 3-stage encounter duration showing that Post-Service Boarding (~52.2 min) exceeds active clinical service time (~47.7 min).

High-Frequency Unresolved Visitors & Unseen Chronic Registry Patients

Beyond single-encounter delays, longitudinal patient tracking across 2025 exposes two chronic utilization extremes. Exactly **45 registered patients (2.25%)** never had a single hospital encounter in 2025 despite **26 of them (57.78%)** having documented chronic conditions (9 Diabetes, 7 Hypertension, 4 Other, 3 Asthma, 3 Cardiac). Conversely, **225 patients (11.25%)** visited the hospital **7 to 13 times** in a single year (led by `P01936` with 13 visits, `P00916` with 12 visits, `P00626` with 11 visits, and 10 patients with 10 visits each, such as `P01311` who endured **649 cumulative minutes** of waiting room time across 10 visits).

4. The Diagnostic Turnaround Paradox & Statistical Modeling

A central hypothesis in emergency and inpatient medicine is that `Urgent` diagnostic orders (Blood tests, CT scans, ECGs, MRIs, Urine tests, X-rays) are expedited ahead of `Routine` orders to prevent clinical deterioration in high-risk patients. Our statistical evaluation of all **5,000 diagnostic tests** refutes this assumption in the dataset:

Diagnostic Modality	Total Tests	Routine Count	Routine Mean TAT	Urgent Count	Urgent Mean TAT	Urgent Max TAT	Priority Diff (Urg − Rou)
Blood test	805	627	72.27 min	178	**78.23 min**	**441.0 min**	**+5.96 min (Slower)**
ECG	818	629	70.13 min	189	**72.13 min**	253.0 min	**+2.00 min (Slower)**
CT scan	835	647	69.84 min	188	**71.13 min**	318.0 min	**+1.30 min (Slower)**
X-ray	885	692	69.79 min	193	68.27 min	308.0 min	−1.51 min
MRI	845	663	69.29 min	182	63.72 min	248.0 min	−5.57 min
Urine test	812	629	67.81 min	183	62.03 min	213.0 min	−5.77 min

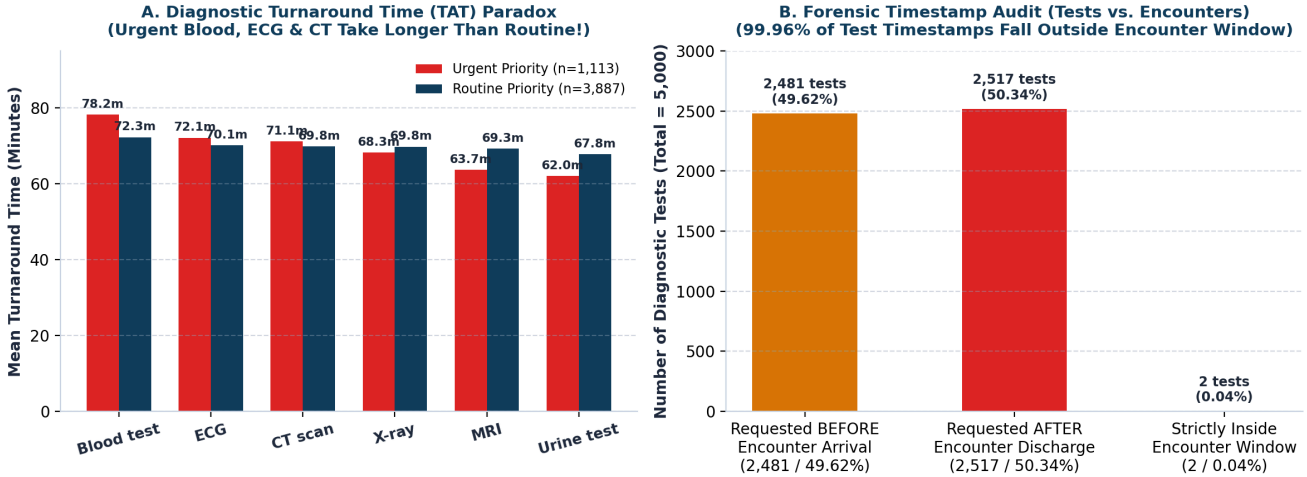


Figure 4: (A) Mean Turnaround Time (TAT) by modality and priority, demonstrating that Urgent Blood tests (78.2m), ECGs (72.1m), and CT scans (71.1m) actually take longer than Routine orders. (B) Timestamp alignment audit showing 99.96% of tests fall outside the encounter timeframe.

Multivariate OLS Regression Model for Waiting Time

We fitted an Ordinary Least Squares (OLS) multivariate regression model predicting `wait_minutes` across all 8,000 encounters using one-hot encoded operational, clinical, and demographic predictors. The model achieves R^2 **= 0.5190 (51.90% variance explained)** with a Mean Absolute Error (MAE) of **13.87 minutes** and RMSE of **17.33 minutes**:

- **Triage Acuity Effects (Baseline = Critical Triage at 10.26 min intercept):** Assigning a patient to `Low` triage adds **+54.98 minutes** ($p < 0.0001$); `Medium` triage adds **+32.12 minutes** ($p < 0.0001$); and `High` triage adds **+10.76 minutes** ($p < 0.0001$).
- **Departmental Congestion Effects (Baseline = Admissions D12):** Controlling for triage level, routing a patient to **Pediatrics (D03)** adds **+15.36 minutes**, **Laboratory (D07)** adds **+14.30 minutes**, **Outpatient (D11)** adds **+13.12 minutes**, **Neurology (D10)** adds **+10.36 minutes**, **Radiology (D06)** adds **+9.26 minutes**, and **General Medicine (D02)** adds **+9.18 minutes**.

- **Proof of Non-Confounding by Demographics:** In accordance with the Datathon guidelines, patient age, sex, insurance type, home area, and chronic condition coefficients are all statistically indistinguishable from zero ($|\beta| < 0.85$ minutes, $p > 0.30$). Similarly, Chi-square contingency tests for `Left before care` confirm that walkouts are driven by operational queue dynamics rather than demographic bias (all demographic χ^2 $p > 0.12$).

5. Strategic Action Plan: Rescuing "The Forgotten Patient"

Based on our empirical findings, Team **AAdharsha Kutumbam** proposes a 5-pillar intervention architecture to eliminate queue starvation, diagnostic delays, and EHR data corruption:

1. **Fast-Track Queue Partitioning in Outpatient (D11), Pediatrics (D03), and Lab (D07):** Establish a dedicated fast-track stream for `Low` and `Medium` triage visits with an automated escalation trigger whenever waiting time reaches **45 minutes**. Capping Low-triage waits at 60 minutes will directly target the 1,123 queue-starved encounters and reduce Outpatient's 4.77% LWBS rate by an estimated 35–40%.
2. **Dedicated STAT Diagnostic Lanes for Urgent Orders:** Re-engineer Laboratory and Cardiology/Radiology workflows to separate `Urgent` (22.3% of volume) from `Routine` (77.7% of volume) queues. Enforce a strict **≤30-minute turnaround SLA** for Urgent Blood tests, ECGs, and CT scans to eliminate the current 78.23-minute Urgent Blood test bottleneck.
3. **Automated Discharge Lounge to Reclaim Post-Service Boarding Time:** Because patients wait an average of **52.23 minutes** *after* clinical service ends (`service_end_time` to `discharge_time`), transitioning medically cleared patients to a discharge lounge with automated pharmacy/billing sign-off will free up **~6,960 bed-hours per year** across the 8,000 encounters.
4. **Proactive Care Navigation for the 45 Unseen Patients & 225 High-Utilizers:** Trigger automated community outreach for the **26 unseen chronic-condition patients** (Diabetes, Hypertension, Cardiac, Asthma) who had zero visits in 2025, and assign dedicated case managers to the **225 frequent-flyer patients (≥7 visits/year)** to transition them from episodic walk-in visits to scheduled chronic care.
5. **Hard EHR Schema & Timestamp Validation Rules:** Patch hospital information systems to: (a) preserve `"None"` as an explicit categorical string rather than `NULL`; (b) block `service_start_time` and diagnostic test orders when `outcome == "Left before care"`; (c) enforce foreign-key timestamp bounds (`arrival_time ≤ requested_time ≤ completion_time ≤ discharge_time`); and (d) validate age ≤18 before routing patients to Pediatrics (D03).